

EduVision_DV

Higher Education Performance Dashboard

Final Project Documentation

Internship	Infosys Springboard 7.0 – Data Visualization
Project	EduVision_DV – Higher Education Performance Dashboard
Primary Tool	Tableau
Data Sources	QS World University Rankings 2024 and Times Higher Education 2024

Prepared for professional internship submission

Executive Summary

EduVision_DV is an interactive higher-education analytics dashboard suite developed as part of the Infosys Springboard 7.0 Data Visualization Internship. The project integrates university information from the 2024 QS World University Rankings and Times Higher Education datasets, applies structured data-cleaning and transformation techniques, engineers six analytical KPIs, and presents the resulting insights through four Tableau dashboards.

The work progressed from an initial combined dataset of approximately 3,466 records and nearly 55 columns to a focused analytical dataset of 620 universities. Particular attention was given to incomplete country information, categorical and demographic fields, percentage and ratio formats, performance scores, and banded ranking values. The final dashboard suite supports university-level, research-level, student-level, and country-level analysis through a consistent interactive interface.

The four dashboards—University Overview, Research Analytics, Student Analytics, and Country Comparison—translate complex ranking and demographic measures into concise visual views suitable for comparative analysis and exploratory decision support.

1. Project Overview

The project objective was to transform raw higher-education ranking data into actionable visual insights through an integrated Tableau dashboard suite. The intended analytical scope covers institutional performance, research output, student demographics, and country-level education trends.

1.1 Project Objectives

- Collect and integrate university performance data from QS and Times Higher Education sources.
- Clean and standardize the combined dataset for analytical and visualization use.
- Engineer six core education KPIs required by the project specification.
- Design and prototype four dashboard modules.
- Develop an integrated Tableau dashboard suite with filters, navigation, and interactive comparisons.
- Present university, research, student, and country-level performance through clear visual analytics.

1.2 Project Scope

Area	Scope
Data	2024 QS World University Rankings and Times Higher Education ranking data
Processing	Integration, cleaning, transformation, missing-data handling, ranking normalization
KPI Engineering	Six project-defined education KPIs
Visualization	Four interactive Tableau dashboards
Interaction	University and Country filters, navigation, comparative views
Final Analytical Dataset	620 universities

2. Requirements and Implementation Alignment

The implementation was organized around the workflow and deliverables defined in the Infosys project blueprint. The table below summarizes the intended modules and the corresponding work completed.

Blueprint Module	Expected Focus	Implemented Work
Data Collection	Collect and integrate QS/THE datasets	Python-based loading and outer merge into a common dataset
Data Cleaning & Transformation	Standardize, normalize and prepare Tableau-ready data	Categorical, demographic, performance-score and ranking transformations
KPI Engineering	Generate six education KPIs	Python KPI-generation script and final Excel dataset
Dashboard Planning	Design four layouts, filters and interactions	Prototype dashboard suite covering all four modules
Dashboard Development	Build required analytical views	Four implemented Tableau dashboards
Dashboard Integration	Global filters, navigation and actions	Consistent navigation and filtering structure across dashboards
Documentation & Delivery	Document sources, KPIs, dashboard guidance and methodology	This project documentation

3. Data Collection and Integration

Two 2024 university-ranking datasets were used as the primary inputs: the QS World University Rankings dataset and the Times Higher Education rankings dataset. The project combined their complementary institutional, performance, ranking, and demographic information.

3.1 Data Integration Process

1. Load the QS and THE CSV datasets using Python and Pandas.
2. Match the QS institution field with the corresponding THE university name field.
3. Perform an outer join so that institutions present in either source are retained in the integrated dataset.
4. Export the merged result as `university_raw_data.csv` for downstream cleaning.

The resulting combined dataset contained approximately 3,466 rows and nearly 55 columns before the detailed cleaning and preparation stage.

3.2 Integration Deliverables

- `data_collection.py`
- `university_raw_data.csv`

4. Data Cleaning and Transformation

Data cleaning was a central part of the project because the two source datasets contained different formats, missing values, ranking conventions, and categorical representations. The cleaning process therefore used field-specific strategies rather than a single blanket rule.

4.1 Country Data Reconciliation

Country information from the QS and THE sources was reconciled using Pandas `combine_first()`. The approach used the available value from one source when the corresponding value in the other source was missing, creating a consolidated `Master_Country` field. The cleaning report records a reduction in country missingness from 56.81% to 0.03%.

4.2 Categorical Missing Values

For categorical fields such as `SIZE`, `FOCUS`, `STATUS`, `RES.`, `AGE`, and `subjects_offered`, missing entries were represented as 'Unknown'. This preserved the record while making the absence of reported information explicit.

4.3 Numeric and Percentage Conversion

Demographic fields containing percentage signs or comma-formatted counts were converted from text to numeric values. This included international-student percentages, international-student ratio measures, student counts, and student-to-staff ratio fields.

4.4 Female-to-Male Ratio Transformation

The `stats_female_male_ratio` field was parsed at the colon separator and the female component was extracted into a dedicated `female_student_pct` field. The project retained the female percentage as the principal diversity metric rather than creating a redundant complementary male-percentage field.

4.5 Demographic Imputation

Missing demographic numerical values were filled using the median of the relevant cleaned field. This choice was intended to provide a representative baseline while reducing the influence of unusually large universities on the imputed value.

4.6 Performance Score Handling

For the project's defined performance-score group, missing score values were represented using zero-fill. This was an explicit project strategy for score fields rather than a general rule applied to every variable.

4.7 Ranking Normalization

Ranking values were standardized so that ties and rank bands could be handled numerically. A tied value such as 116= was converted to 116, while a band such as 1201–1400 was represented using its midpoint, 1300.5. This enabled consistent numerical sorting and charting.

4.8 Missing Ranking Values

Missing ranking values were intentionally retained as NaN rather than deleting the associated universities or assigning an artificial large rank. This allowed records without ranking information to remain available for geographical and demographic analysis while ranking-specific visuals could naturally exclude unavailable values.

4.9 Final Cleaning Output

Following the cleaning and preparation process, the working analytical dataset contained 620 universities. The cleaned data was exported without the Pandas index so that it could be used directly by the visualization layer.

Stage	Result
Initial source integration	QS + THE datasets
Integrated dataset	≈3,466 rows; nearly 55 columns
Cleaning & transformation	Field-specific standardization and missing-data handling
Final analytical dataset	620 universities
Visualization-ready output	university_final_dataset.xlsx

5. Education KPI Engineering

Six core KPIs were engineered from the cleaned dataset to provide consistent analytical measures across the dashboard suite. The calculations were implemented in Python and the resulting KPI fields were rounded to two decimal places before export.

KPI	Definition / Construction
Global Ranking Score	Row-wise mean of the QS Overall SCORE and THE scores_overall values after numeric cleaning.
Research Impact Score	Row-wise mean of QS Citations per Faculty Score and THE scores_citations.
Faculty-to-Student Ratio	Cleaned stats_student_staff_ratio field.
International Student Percentage	Cleaned stats_pc_intl_students field.
Academic Reputation Score	Cleaned Academic Reputation Score field.
Research Productivity Index	Row-wise mean of THE scores_research and QS International Research Network Score.

The KPI-generation script exports the prepared result as university_final_dataset.xlsx, which serves as the analytical input for the dashboard layer.

6. Dashboard Planning and Prototyping

A prototype dashboard set was prepared before the final Tableau implementation. The prototype established a common navigation pattern, KPI-card approach, filter placement, and analytical layout for the four required modules.

6.1 Prototype Modules

- University Overview
- Research Analytics
- Student Analytics
- Country Comparison

The prototype was used as a planning artifact. The final dashboards evolved from the planned concepts to reflect the cleaned 620-university dataset and the KPI fields actually available for analysis.

7. Final Tableau Dashboard Suite

The final dashboard suite is organized into four interconnected analytical views. Each dashboard combines KPI cards, charts, geographic views and comparative visuals while maintaining a consistent visual language and filter structure.

7.1 University Overview

Provides an institution-level view of overall university performance. The dashboard includes total universities, average overall score, employer reputation and academic reputation KPIs, together with top global rankings, employer reputation leaders, country distribution, performance trends, academic reputation analysis and institutional comparison.

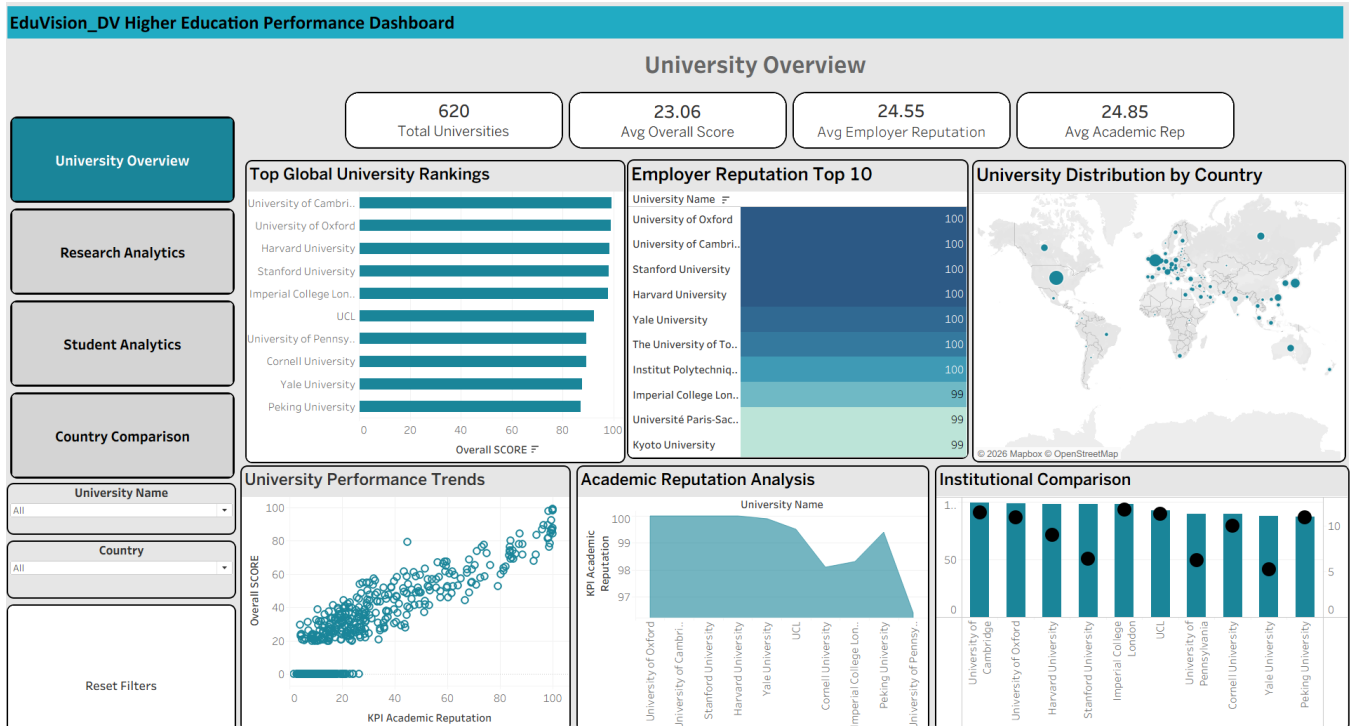


Figure 7.1. Implemented Tableau dashboard — University Overview

7.2 Research Analytics

Focuses on research-oriented performance indicators, including average research impact, citations per student, international research network and research productivity. Supporting visuals cover publication analysis, top research institutions, productivity trends, citation leaders, citation performance and research-impact comparison.

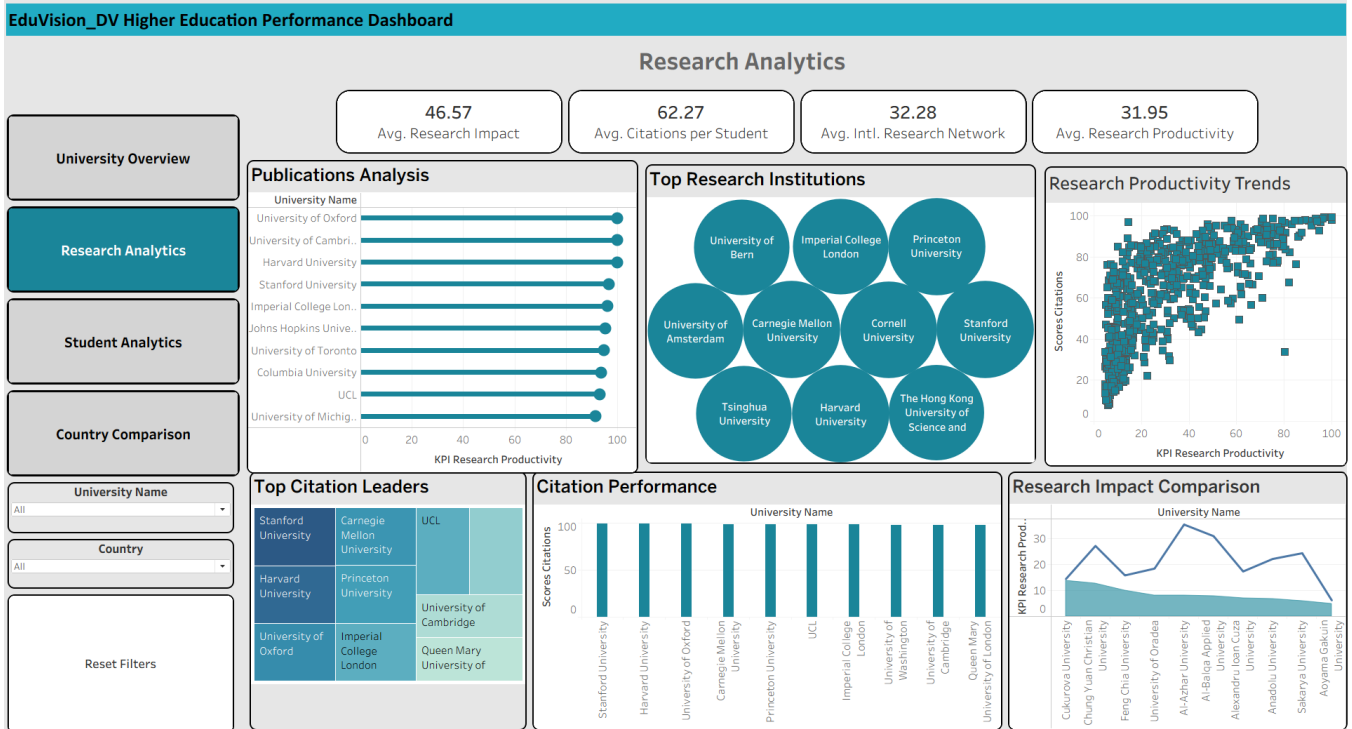


Figure 7.2. Implemented Tableau dashboard — Research Analytics

7.3 Student Analytics

Examines the student dimension of higher education through international-student percentage, female-student percentage, faculty ratio and total-student KPIs. Visuals cover international-student analysis, student distribution, employment outcomes, student diversity, enrollment comparisons and faculty-to-student ratio analysis.

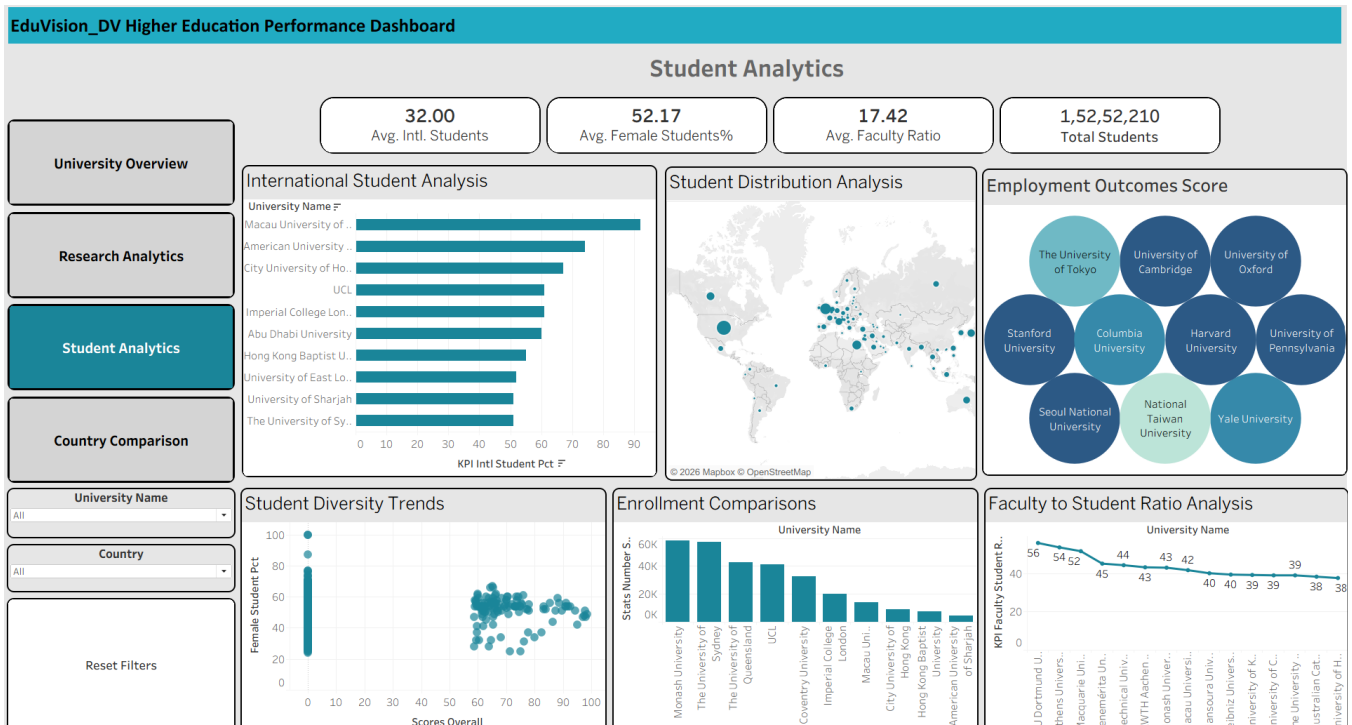


Figure 7.3. Implemented Tableau dashboard — Student Analytics

7.4 Country Comparison

Provides a country-level view of higher-education performance. It combines country counts, average global score and international outlook with top-performing countries, international-outlook spread, regional trends, education-performance benchmarking, country ranking comparison and top student populations.

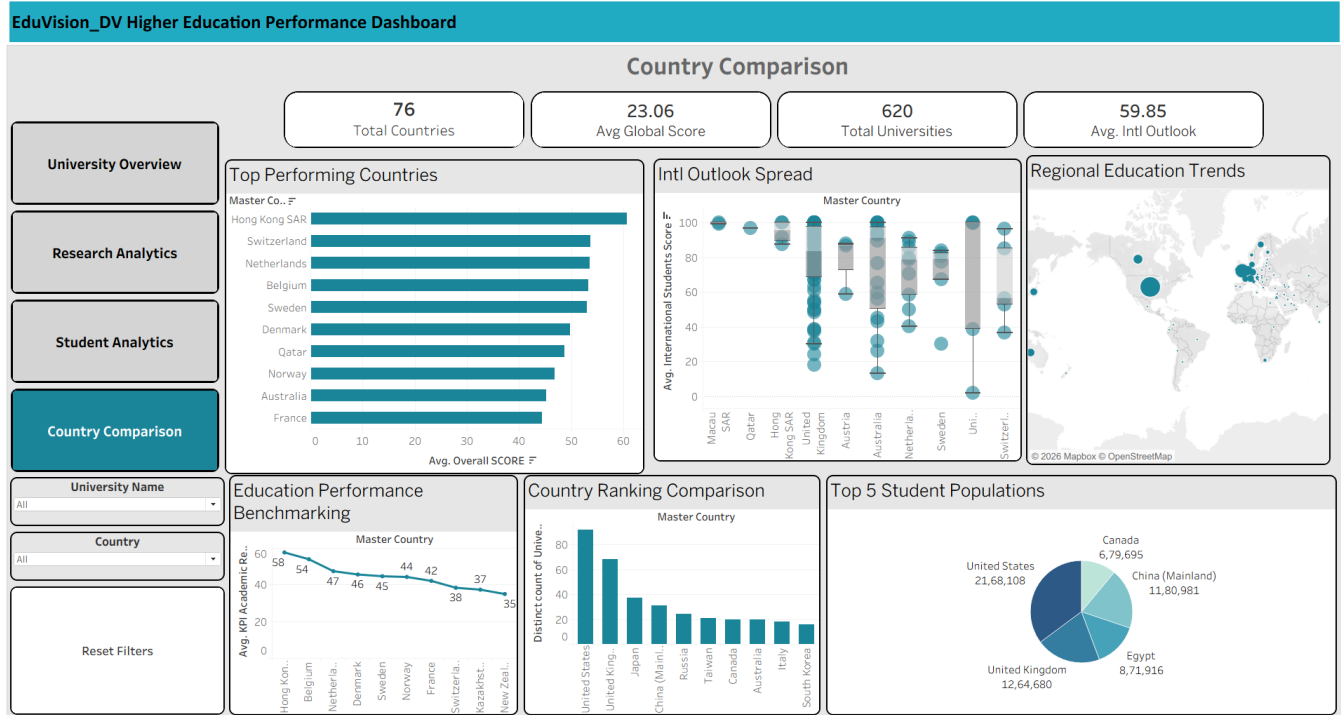


Figure 7.4. Implemented Tableau dashboard — Country Comparison

8. Technical Implementation and UI/UX

The dashboards were designed as a consistent analytical interface rather than as four isolated sheets. The implementation uses tiled layout containers, deliberate spacing, consistent typography, and a restrained teal/grey visual theme.

Design Area	Implementation
Layout	Grid-based tiled structure using horizontal and vertical containers.
Spacing	Custom padding and negative space to separate analytical sections and improve visual hierarchy.
Visual consistency	Common teal/grey palette, KPI-card treatment, titles and navigation structure.
Navigation	Direct navigation between University Overview, Research Analytics, Student Analytics and Country Comparison.
Filtering	University Name and Country controls used to support focused exploration.
Comparative analysis	Charts and KPI views designed to compare institutions, countries and performance dimensions.
Tableau architecture	Attention to container hierarchy and rendering behavior during dashboard construction.

9. Dashboard User Guide

5. Open the dashboard suite and begin with University Overview for a high-level institutional view.
6. Use the University Name or Country filter to narrow the analytical context.
7. Move to Research Analytics to examine publications, citations, research productivity and research impact.
8. Use Student Analytics to explore international students, diversity, enrollment and faculty-to-student relationships.
9. Use Country Comparison for country-level benchmarking and regional education patterns.
10. Clear selections using the reset-filter control when returning to the overall dataset view.

10. Education Analytics Methodology

The analytical methodology follows a layered approach: source integration first establishes a common institutional dataset; cleaning then makes the fields analytically usable; KPI engineering creates standardized measures; and Tableau converts those measures into interactive visual narratives.

Layer	Purpose
Source layer	Capture complementary university-ranking and demographic information from QS and THE.
Integration layer	Combine source records while retaining institutions present in either dataset.
Preparation layer	Resolve formats, missing values, categorical inconsistencies and ranking bands.
KPI layer	Create six standardized measures for cross-dashboard analysis.
Visualization layer	Present institutional, research, student and country-level patterns.
Interaction layer	Allow users to filter and compare the prepared data through the dashboard suite.

11. Key Challenges and Solutions

Challenge	Approach Used
High and uneven missingness across source fields	Applied field-specific strategies instead of one universal imputation rule.
Incomplete country information	Cross-referenced QS and THE country fields to create Master_Country.
Text-formatted percentages and counts	Removed formatting characters and converted fields to numeric types.
Female:Male ratio stored as text	Extracted the female component into female_student_pct.
Banded and tied rankings	Converted rank strings to numeric values and used midpoints for ranges.
Missing ranks	Retained NaN so unranked records remained available for non-ranking visuals.
Dashboard container restrictions and rendering issues	Used tiled containers and adjusted spacing/layout to maintain a stable dashboard structure.

12. Project Deliverables

Deliverable	Purpose
data_collection.py	Loads and integrates the QS and THE source datasets.
university_raw_data.csv	Merged raw dataset before detailed cleaning.
education_cleaning.ipynb	Documents the cleaning and transformation workflow.
cleaned_university_data_2024.csv	Cleaned analytical dataset.
generate_education_kpis.py	Calculates the six core education KPIs.
university_final_dataset.xlsx	Final KPI-enriched dataset used for visualization.
Dashboard prototype PDF	Planning/prototyping artifact for the four dashboard modules.
Final Tableau dashboard suite	Four implemented dashboards: University Overview, Research Analytics, Student Analytics and Country Comparison.
Final project documentation	Consolidated technical and analytical documentation.

13. Testing and Validation

The project specification called for validation of KPI calculations, ranking calculations, dashboard interactions and educational metrics. In the supplied project material, no separate quantitative QA checklist or measured accuracy report was provided. Accordingly, this document records the validation scope without inventing test results.

- KPI calculation logic was implemented in the dedicated KPI-generation script.
- Ranking fields were normalized to support numerical sorting and dashboard use.
- Dashboard layouts and interactions were developed around the prepared dataset and defined filter/navigation structure.
- The final documentation preserves the distinction between the project requirements and the implementation evidence supplied.

14. Learning Outcomes and Competencies

- Data integration using Python and Pandas.
- Practical handling of missing, inconsistent and text-formatted data.
- Transformation of ranking and demographic fields into analysis-ready variables.
- KPI design and implementation for higher-education analytics.
- Tableau dashboard design, layout management and interactive filtering.
- Translation of complex datasets into concise visual narratives.
- Troubleshooting dashboard rendering and container-hierarchy constraints.

15. Conclusion

EduVision_DV demonstrates an end-to-end data-visualization workflow in which raw higher-education ranking data was integrated, cleaned, transformed, enriched with analytical KPIs, and

presented through an interactive Tableau dashboard suite. The project resulted in a focused 620-university analytical dataset and four complementary dashboards covering institutional performance, research, students and country-level benchmarking.

The principal value of the project lies in connecting rigorous data preparation with clear visual communication. The resulting dashboard suite provides a structured way to explore higher-education performance while maintaining a consistent and professional user experience.

Appendix A — Principal Data Fields Used in Analysis

Field / Group	Analytical Role
University_Name	Institution identifier and chart label.
Master_Country	Geographic grouping and country filtering.
Overall SCORE / scores_overall	Inputs to Global Ranking Score.
Academic Reputation Score	Academic reputation analysis.
Employer Reputation Score	Employer-oriented institutional comparison.
Citations per Faculty Score / scores_citations	Research impact and citation analysis.
scores_research	Research performance input.
International Research Network Score	Research collaboration/productivity input.
stats_student_staff_ratio	Faculty-to-student analysis.
stats_pc_intl_students	International-student analysis.
female_student_pct	Student diversity analysis.
stats_number_students	Enrollment and student-population analysis.
scores_international_outlook	Country/international outlook analysis.
2024 RANK / rank	University ranking analysis.

Appendix B — Reference to Project Specification

The project structure, four-dashboard requirement, six KPI requirement, expected deliverables and technology stack were aligned with the supplied Infosys Springboard project blueprint. The blueprint identifies Tableau for visualization and Pandas/NumPy/Python for data processing and specifies documentation covering dataset sources, KPI definitions, dashboard guidance and education-analytics methodology.